

Embodied At Sea: Behavior Cloning for Marine Archaeology and Wildlife Conservation

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<https://lair-research-nyu.github.io/Embodied-at-Sea/>

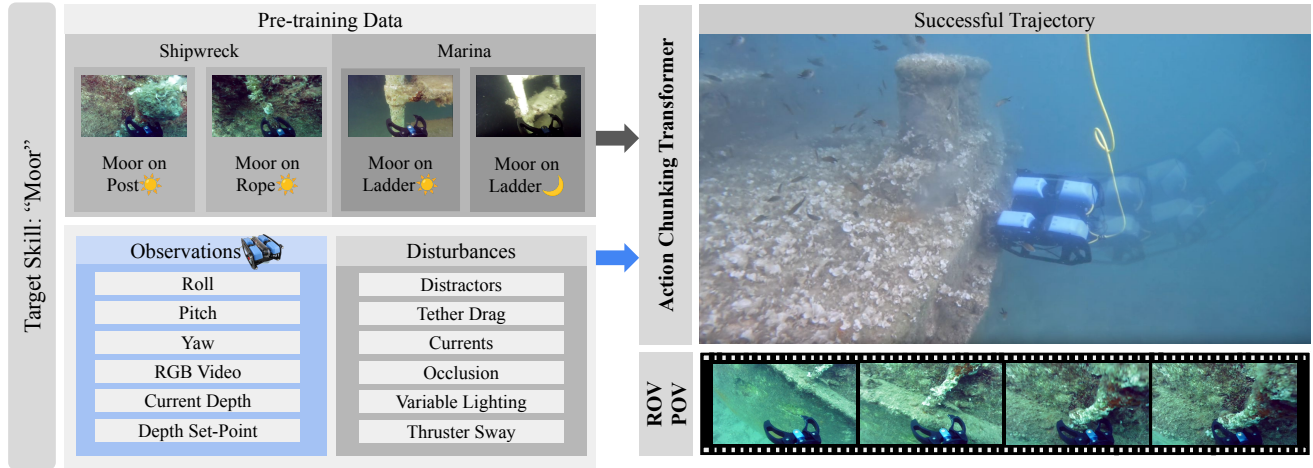


Fig. 1: **Overview of Autonomous Underwater Mooring using Action Chunking Transformers (ACT).** (Left) Pre-training data collected across shipwreck and marina environments incorporating multi-modal observations and real-world disturbances. These include marine life distractors, tether drag, ocean currents, visual occlusion from water particulate, variable lighting based on depth or time of day, and thruster sway inherent to fluid dynamics. (Right) Autonomous policy execution showing closed-loop mooring trajectory on the *Coralita XI27* shipwreck.

Abstract—Modern behavior cloning (BC) architectures have demonstrated promising generalization capabilities for robotic manipulation in terrestrial environments; however, their deployment and evaluation in underwater settings remain largely unexplored. This work presents preliminary results from an ongoing case study evaluating both a single-task (Action Chunking Transformer) and a generalist Vision-Language-Action policy (SmolVLA) for two tasks in marine archaeology and conservation: grasping and mooring. Experiments are performed with a BlueROV2 on and around the WWII wreck *Coralita XI27* and at Manoel Island Marina, Malta. Across 748 collected episodes and 508 gradable on-hardware rollouts, the single-task ACT policy substantially outperformed SmolVLA (47.9% vs. 7.9% pooled success rate). These results underscore the severe domain shift posed by marine visual distortions and dynamic currents, revealing key bottlenecks in adapting current generalist architectures to underwater field manipulation.

Index Terms—Behavior Cloning, Field Robotics, Remotely Operated Vehicle (ROV), Marine Archaeology and Conservation, Vision Language Action Models (VLA), Action Chunking Transformers (ACT)

I. INTRODUCTION

Recent advances in Behavior Cloning (BC) [1] have enabled robust single-task policies from relatively little data.

While limited prior works explore the performance of these approaches in aquatic robotics [2], [3], deployment in real marine archaeology and conservation missions remains largely unexplored.

Marine archaeologists routinely perform tasks such as environmental monitoring and specimen retrieval, constrained by poor visibility, dynamic currents, and the physiological limits of human divers. At deeper sites, monitoring requires repeated dives that are slow, costly, and risky due to decompression limits [4]. Automating these operations could accelerate scientific research while reducing diver risk.

BC-based policies, particularly Vision-Language-Action (VLA) models, have shown remarkable generalization to novel tasks and objects, making them an appealing candidate for underwater manipulation. However, they are trained almost exclusively on terrestrial datasets featuring stable lighting and fixed-base manipulators, while underwater vehicles operating in marine environments experience unconstrained 6-DOF motion, drag and turbidity, visual distortion and occlusion, and distractors (see Fig. 1). Whether these policies retain their generalization capabilities under such conditions and across diverse archaeological sites remains an open question.

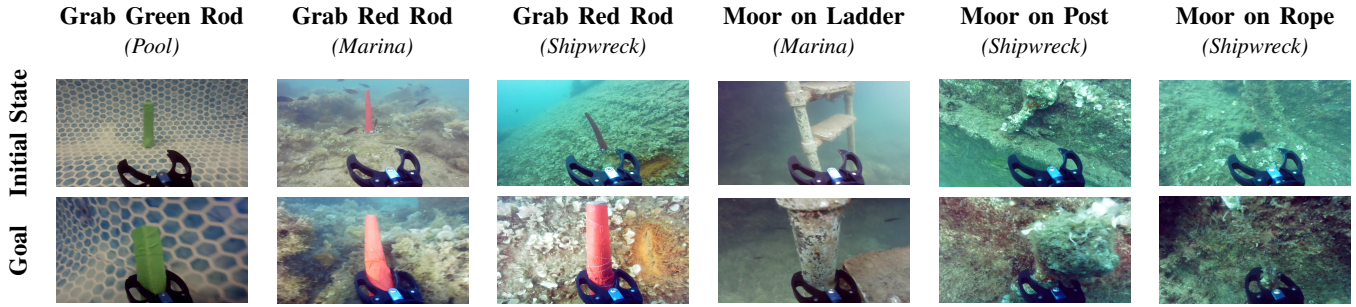


Fig. 2: **Experimental tasks evaluated across different underwater environments.** Data collection in the Pool (Control) was fully illuminated with no external currents applied to the robot. Data collection at the Marina and Coralita X127 shipwreck took place during both day and night, featuring varying disturbances (see Fig. 1).

This work presents the following core contributions:

- 1) A real-world case study of BC manipulation models deployed on an underwater Remotely Operated Vehicle (ROV) at a Mediterranean shipwreck for archaeology and conservation tasks.
- 2) An empirical evaluation of BC model performance under varying lighting, visibility, and environmental disturbances.
- 3) Analysis of failure modes when transferring BC models underwater, and key directions for the development of robust marine embodied intelligence.

II. RELATED WORK

Foundation models for robotic manipulation have enabled increasingly generalizable robotic behaviors on terrestrial tasks. Action Chunking Transformers (ACT) [1] learns complex manipulation from low-cost demonstrations, Diffusion Policies [5], SmolVLA [6], and π_0 [7] extends this via large-scale pretraining, and DynamicVLA [8] improves inference efficiency for dynamic tasks by overlapping inference and execution across action chunks.

Other works adapt VLAs beyond their training distribution: AirVLA [9] adjusts actions via a physics-aware, embodiment-specific module, UMI-Underwater [3] adapts land-trained diffusion policies underwater with a depth-based affordance predictor, and AquaBot [2] shows learning-based policies can improve through underwater data collection. Still, traditional control approaches offer precision but not cross-site generalization, and scaling foundation models to handle unpredictable hydrodynamic disturbances without full retraining remains a critical challenge.

III. METHODOLOGY

In order to examine the performance of BC-based policies in field-realistic conditions, two task families were explored (see Fig. 2): *grab*, approaching an object and grasping it with the end-effector to simulate sampling a specimen, and *moor*, approaching a stationary post or rope and latching on with the end-effector to simulate a docking procedure.

Experiments used a BlueROV2 (a vector-configured mini-ROV with six T200 thrusters) with the Newton Gripper end-effector installation, controlled via a Raspberry Pi 4 companion computer with visual feedback from a front-facing

1080p wide-angle camera. In total, we collected 748 human-teleoperated episodes across three sites, progressing from a controlled test pool (79 episodes) to the *Coralita X127* shipwreck (335 episodes total) and finally to Manoel Island Marina, Malta (334 episodes total). To improve dataset robustness, our collection pipeline includes recovery trajectories obtained by resampling directly from failure states.

We trained one ACT and one SmolVLA policy per task family and evaluated both on-hardware. All four models take in a six-dimensional observational space (see Fig. 1) and outputs a six-dimensional action space: horizontal velocity (x, y), yaw rate (ω_z), a vertical altitude setpoint (z_{ref}) with z_{ref} tracked by a closed-loop PID controller for stable depth-following, camera tilt toggle, and the end-effector setpoint.

IV. RESULTS AND DISCUSSION

Our results can be summarized by Table I, which reports success by task and location over 508 gradable autonomous policy rollouts.

To test whether a single trained policy could generalize across physically distinct instances of the same task family, we evaluated one fixed ACT checkpoint on three mooring variants without any per-site fine-tuning: on a ladder at the Marina, on a tethered rope on the Shipwreck, and on a post on the Shipwreck (see Fig. 2). Success rates were 56.3% ($n=206$) on the ladder, 66.7% ($n=21$) on rope, and 36.8% ($n=19$) on the mooring post (Table I). The comparable success rates on the ladder and rope tasks suggest that the policy can transfer its mooring strategy across physically distinct sites without per-site fine-tuning. The lower success rate observed for the post likely results from its less forgiving target geometry compared to the ladder and rope, even though both the rope and post exhibited greater occlusion from algae and marine overgrowth than the ladder.

ACT consistently outperformed SmolVLA on every task family (Table I) pooled across sites: 37.6% versus 0.0% on *grab* and 55.3% versus 12.5% on *moor*. The gap is largest on the tasks with the most training data: at the Marina ladder, ACT reached 56.3% on mooring versus SmolVLA's 12.5%, and 42.9% versus 0.0% on *grab red rod*, where SmolVLA never succeeded. ACT's best *grab* result was *grab green rod*

TABLE I: **Success rate by task and location.** ACT consistently outperforms SmolVLA, and Mooring is the best-performing task family for both architectures.

Task	Location	Policy	n	SR (%)	95% CI
Grab (red rod)	Shipwreck	ACT	150	33.3	[26.3, 41.2]
Grab (red rod)	Marina	ACT	14	42.9	[21.4, 67.4]
Grab (red rod)	Marina	SmolVLA	28	0.0	[0.0, 12.1]
Grab (green rod)	Pool	ACT	22	63.6	[43.0, 80.3]
Moor (ladder)	Marina	ACT	206	56.3	[49.5, 62.9]
Moor (ladder)	Marina	SmolVLA	48	12.5	[5.9, 24.7]
Moor (rope)	Shipwreck	ACT	21	66.7	[45.4, 82.8]
Moor (post)	Shipwreck	ACT	19	36.8	[19.1, 59.0]

in the pool (63.6%), which is expected given the controlled environment.

When pooled across all tasks and locations, ACT succeeded on 47.9% of rollouts and demonstrated strong recovery trajectories during trials, versus 7.9% for SmolVLA which demonstrated no recovery capabilities. We hypothesize this ability gap reflects both architecture and data regime. ACT produces short, temporally coherent action chunks that are replanned frequently, keeping execution close to closed-loop. This may provide comparative robustness to disturbances that are largely absent from the terrestrial data SmolVLA’s backbone was pretrained on. SmolVLA, a substantially heavier model, both replans more slowly and was trained on far less task-specific underwater data than our ACT model, so the performance gap likely reflects inference latency and data quantity rather than a fundamental architectural ceiling.

Both tasks evaluated across daylight variations show a consistent performance drop at night. The *moor on ladder* success rate fell from 63.9% ($n=97$) in the daytime to 49.5% ($n=109$) at night, and the *grab red rod* success rate fell from 42.4% ($n=66$) to 26.2% ($n=84$) respectively. The drop was larger for grasping, which is consistent with the grasping trajectories being less forgiving to disturbances: unlike mooring, which tolerates vertical error since the ROV can latch anywhere along the ladder or post, grasping the small, swaying rod leaves little margin for error, especially with the reduced contrast and noise of nighttime lighting.

V. CONCLUSIONS AND FUTURE WORK

Our results expose two compounding obstacles for BC-based underwater field manipulation:

- 1) **A pretrained vision backbone mismatched to the domain.** SmolVLA’s value comes from a backbone pretrained on large-scale terrestrial data. Freezing the backbone leaves every underwater observation out-of-distribution and degrades feature quality, whereas fine-tuning it on our comparatively small, noisy dataset overfits and erodes the broad representations that motivated using a VLA in the first place. ACT’s much smaller vision encoder can instead be trained end-to-end on in-domain demonstrations, trading broader generalization for stronger single-task performance.
- 2) **Persistent disturbance under a slow policy.** Unlike a fixed-base manipulator, ROVs are continuously jostled

by waves and currents across all six degrees of freedom, shifting the camera viewpoint and the vehicle’s pose relative to the target between consecutive actions. As a result, a stationary target manipulation object in the world frame appears to move in the robot frame. This turns an otherwise static task into a dynamic one that penalizes any policy too slow to react. SmolVLA, a heavier model, replans infrequently, so a chunk conditioned on a stale observation drifts before the next inference step. ACT’s lighter architecture replans faster, staying closer to closed-loop while sacrificing the multi-task chaining capabilities enabled in VLAs.

Future work follows from these obstacles: Extending the observation space with IMU data would allow BC policies to account for ROV motion induced by hydrodynamic disturbances, reframing underwater manipulation as a partially dynamic task rather than a static one. Improving chunked inference speed, as explored in DynamicVLA, would allow heavier VLAs to replan quickly enough to track disturbances in near real-time. Finally, collecting additional trials across *Coralita X127* and other wrecks, evaluating longer-horizon tasks such as tether untangling, and progressively extending deployments to greater depths would allow policies to advance toward robust underwater embodied intelligence, ultimately expanding robotic access to archaeological sites that remain challenging for human divers.

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REFERENCES

- [1] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, “Learning fine-grained bimanual manipulation with low-cost hardware,” 2023. [Online]. Available: <https://arxiv.org/abs/2304.13705>
- [2] R. Liu, H. Ha, M. Hou, S. Song, and C. Vondrick, “Self-improving autonomous underwater manipulation,” 2024. [Online]. Available: <https://arxiv.org/abs/2410.18969>
- [3] H. Li, L. Y. Chung, J. Goler, R. Zhang, X. Xie *et al.*, “Umi-underwater: Learning underwater manipulation without underwater teleoperation,” 2026. [Online]. Available: <https://arxiv.org/abs/2603.27012>
- [4] E. Gouin, E. Dugrenot, and B. Gardette, “Extremely deep bounce dives: planning and physiological challenges based on the experiences of a sample of french-speaking technical divers,” *Diving and Hyperbaric Medicine*, vol. 55, no. 2, p. 203, 2025.
- [5] C. Chi, Z. Xu, S. Feng, E. Cousineau, Y. Du *et al.*, “Diffusion policy: Visuomotor policy learning via action diffusion,” 2024. [Online]. Available: <https://arxiv.org/abs/2303.04137>
- [6] M. Shukor, D. Aubakirova, F. Capuano *et al.*, “Smolvla: A vision-language-action model for affordable and efficient robotics,” 2025. [Online]. Available: <https://arxiv.org/abs/2506.01844>
- [7] K. Black, N. Brown, D. Driess, A. Esmail, M. Equi, C. Finn *et al.*, “ π_0 : A vision-language-action flow model for general robot control,” 2026. [Online]. Available: <https://arxiv.org/abs/2410.24164>
- [8] H. Xie, B. Wen, J. Zheng *et al.*, “Dynamicvla: A vision-language-action model for dynamic object manipulation,” 2026. [Online]. Available: <https://arxiv.org/abs/2601.22153>
- [9] J. Tucker, D. Liu, A. Swann *et al.*, “ π , but make it fly: Physics-guided transfer of vla models to aerial manipulation,” 2026. [Online]. Available: <https://arxiv.org/abs/2603.25038>